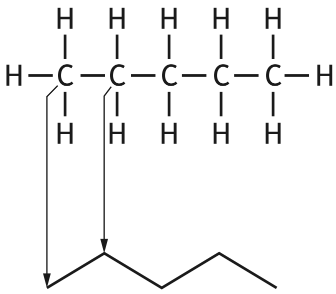
12. Chemists can represent the structures of carbon-chain molecules in a simplified way known as skeletal formulae.

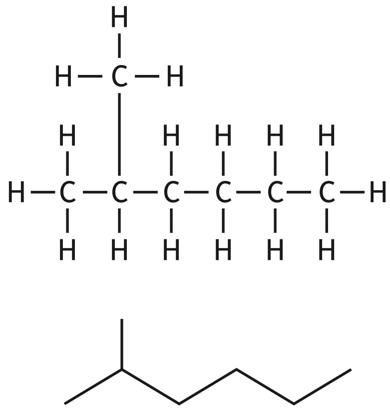
In a skeletal formula, the symbols for carbons, and for the hydrogens attached to carbons, are not shown. The carbon-chain structure is shown as a zig-zag line in which each bend represents a carbon atom. The ends of the zig-zag line also represent carbon atoms.

Each carbon atom is understood to have the correct number of hydrogen atoms attached.

Some examples are shown below.

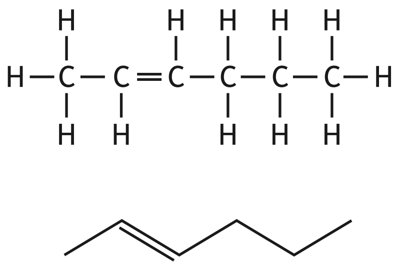


pentane



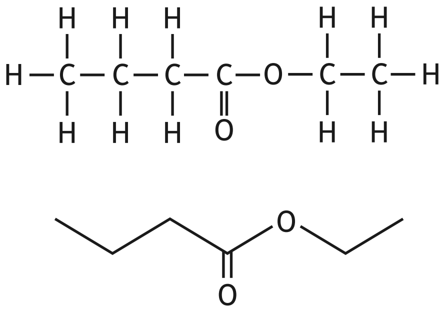
2-methylhexane

A double bond is shown by a double line in the zig-zag line.

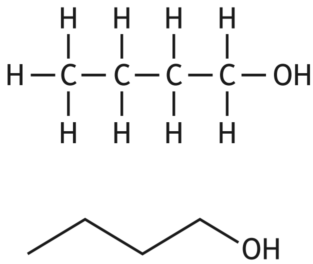


hex-2-ene

Other atoms, such as oxygen, are shown by their symbols attached to the appropriate part of the zig-zag line.



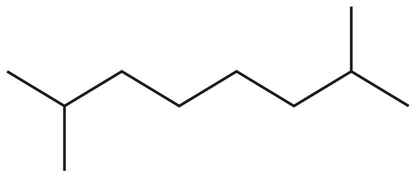
ethyl butanoate



butan-1-ol

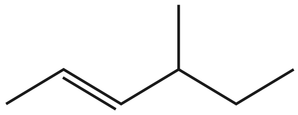
12. **(continued)**

(a) A skeletal formula is shown.



Name this molecule. **1**

(b) Draw the full structural formula for the molecule shown below. **1**



(c) Draw the skeletal formula for pentanoic acid. **1**

[END OF QUESTION PAPER]